

- 7 X-rays are produced when electrons accelerated by a large electrical potential difference impinge upon a metal target. The X-ray spectrum of copper shown in Fig. 7.1 is produced by bombarding a copper target with high-energy electrons. The spectrum consists of two main components: a continuous spectrum (bremsstrahlung) and a line spectrum (characteristic X-rays).

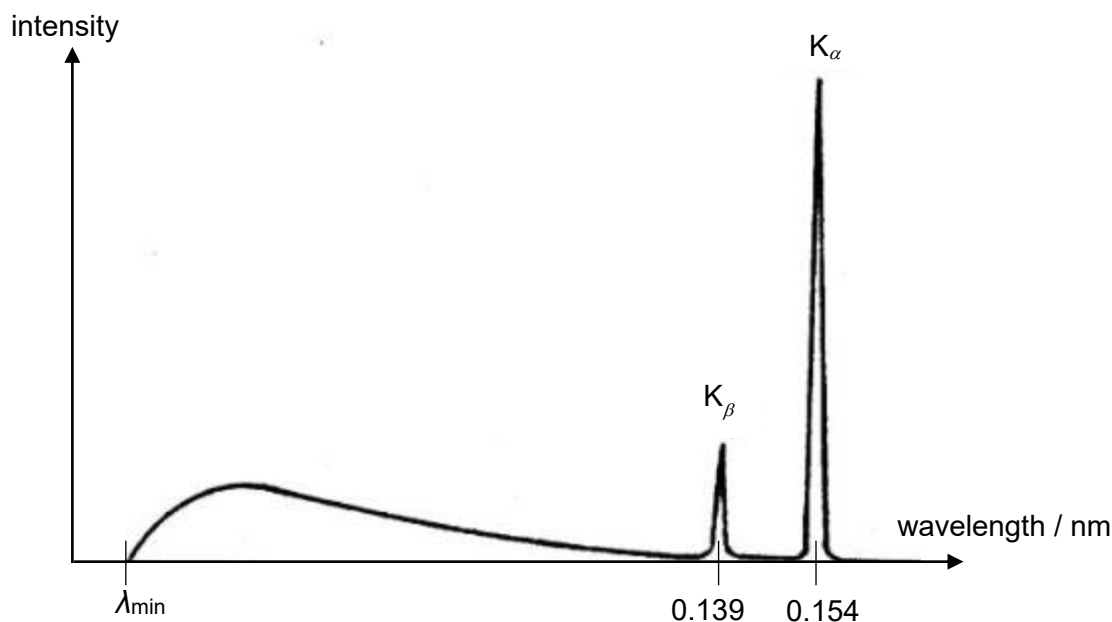


Fig. 7.1

(a) Explain the shape of:

(i) the continuous spectrum,

The bombarding electrons can lose any fraction of their initial kinetic energy, up to the maximum in a single interaction.[1] As a large number of electrons lose differing amounts of kinetic energies, X-ray photons of various wavelengths are emitted, resulting in a continuous X-ray spectrum represented by the solid line. [1].....[2]

(ii) the sharp peaks in the spectrum.

When a fast bombarding electron knocks out an inner shell electron from a target atom, a vacancy is created. An electron in the outer shell de-excites to fill the vacancy, emitting an X-ray photon. [1] X-ray photons emitted in such de-excitations have energies that are equal to the energy differences between the energy levels of the two shells, hence producing X-rays of specific frequencies that appear as sharp peaks in the spectrum. [1].....[2]

- (b) (i) Calculate the energy of a K_α photon.

K_α photons have lower energy (and hence longer wavelength) than K_β photons.

$$\lambda = 0.154 \text{ nm}$$

$$E = \frac{hc}{\lambda} = \frac{6.63 \times 10^{-34} \times 3.0 \times 10^8}{0.154 \times 10^{-9}}$$

$$E = 1.29 \times 10^{-15} \text{ J} = \underline{8070 \text{ eV}}$$

energy = eV [1]

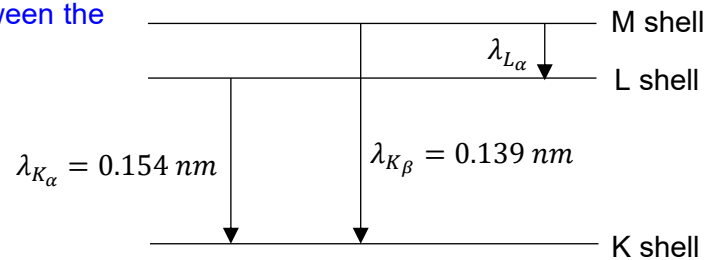
- (ii) L_α photons are emitted when inner shell electrons de-excite from the M shell to the L shell. Calculate the wavelength of the L_α photon.

Comparing energies of photons emitted between the K, L and M shells, we see that:

$$\frac{hc}{\lambda_{K\beta}} = \frac{hc}{\lambda_{K\alpha}} + \frac{hc}{\lambda_{L\alpha}}$$

$$\frac{1}{0.139} = \frac{1}{0.154} + \frac{1}{\lambda_{L\alpha}}$$

$$\lambda_{L\alpha} = 1.4 \text{ nm}$$



wavelength of L_α line = nm [2]

- (c) The minimum wavelength $\lambda_{\min}$ observed in the continuous spectrum depends on the accelerating voltage V applied to the electrons. If the accelerating voltage is 30 kV, calculate $\lambda_{\min}$.

Explain your working.

X-rays of wavelength $\lambda_{\min}$ are emitted if all of the electron's KE is converted into a single photon:

Energy of photon = KE of bombarding electron

$$\frac{hc}{\lambda_{\min}} = q\Delta V$$

$$\frac{6.63 \times 10^{-34} \times 3.0 \times 10^8}{\lambda_{\min}} = 1.6 \times 10^{-19} \times 30,000$$

$$\lambda_{\min} = 4.1 \times 10^{-11} = \underline{0.041 \text{ nm}}$$

$\lambda_{\min} = \dots\dots\dots$ nm [2]

- (d) Explain why knowledge of the X-ray spectra of elements like copper can be used to identify the existence of these atoms in materials.

When such atoms within materials are excited (by electron bombardment), they will emit X-rays with energies equal to the difference between the higher and lower energy levels. As these energy levels are specific to each element, so the energies and wavelengths of characteristic peaks are also unique to each element. Thus, x-ray analysis can be performed on an unknown target to determine the elements present. [1]

[Total: 10]

